

Name: _____

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Class: _____

MyDesign Portfolio

Element E Initial Template

How to use this template

Students: The template below is one way to describe how your proposed solution is supported by STEM concepts and principles, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit, be sure to delete the instructions.

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

This section should describe how your proposed solution is supported by STEM (science, technology, engineering, mathematics, computer science, etc.) concepts.

Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Go beyond describing the engineering design process to describe the STEM concepts as they apply to your proposed solution in order to meet the design requirements.

Be sure to apply the applicable STEM concepts in context, directly to the specifics of your design. Review concepts learned in your other classes and tackle new concepts as appropriate. Include any relevant calculations.

Element C indicates that you should have at least three high-priority design requirements to receive a score of 3 (and four for a score of 4, and five for a 5). Your number of design requirements and your number of high-priority design requirements should be the same throughout your portfolio. You should use the same three to five design requirements here too, and apply at least three of them to STEM concepts to achieve a score of 5 on E1. Use as many copies of this sub-template as needed.

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STEM Principle for High-Priority Design Requirement #1

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your design:

STEM Principle for High-Priority Design Requirement #2

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your design:

STEM Principle for High-Priority Design Requirement #3

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your design:



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STEM Principle for High-Priority Design Requirement #4 If you only have 3 Design Requirements you can delete this one.

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your design:

STEM Principle for High-Priority Design Requirement #5 If you only have 3 Design Requirements you can delete this one.

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your design:

Proposed Solution to STEM Principles and Practices (E2)

While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.

For E2, use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem. Use as many copies of this sub-template as needed.



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STEM Principle for Design Solution #1

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:

STEM Principle for Design Solution #2 - Iteration of the first design

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:

STEM Principle for Design Solution #3 - Iteration of the second design

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:



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Evidence Design is Reviewed (E3)

"Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

E3 is about experts reviewing what you've said about the STEM connections in E1 and E2. It's important here to talk to an appropriate field expert who knows about the STEM content of the project - not to a stakeholder. We need to know what the expert said. Document in this sub-element what the expert said about the connections you've made from your design requirements to STEM principles and practices (E1) and what they said about the ways STEM principles and practices substantiate your design solution (E2). This documentation will support you to give a corrective response in E5. Potential formats for review include: interview notes, emails, audio/video tape with transcript, etc. Use as many copies of this sub-template as needed.

Documentation #1:

- What format: Interview notes, emails, audio/video tape with transcript, etc.
- What the expert said about the connection you've made from your design requirements to STEM principles and practices:

- What the expert said about how STEM principles and practices substantiate your design solution:

Documentation #2:

- What format: Interview notes, emails, audio/video tape with transcript, etc.
- What the expert said about the connection you've made from your design requirements to STEM principles and practices:



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- What the expert said about how STEM principles and practices substantiate your design solution:

Documentation #3:

- What format: Interview notes, emails, audio/video tape with transcript, etc.
- What the expert said about the connection you've made from your design requirements to STEM principles and practices:

- What the expert said about how STEM principles and practices substantiate your design solution:

Experts with Credentials (E4)

Document who your field experts are (names are not necessary) and what qualifies them as an expert. Use as many copies of this sub-template as needed.

Expert #1: Describe credential(s) and what makes them qualified as an expert.

Expert #2: Describe credential(s) and what makes them qualified as an expert.

Expert #3: Describe credential(s) and what makes them qualified as an expert.

Verification or Corrective Response to Design Review (E5)

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses. For example, "_____ told us



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_____, which informed our decision to change/reconsider _____. " (or "...which verified/reinforced our decision to _____."))

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response:

Interview with Expert #2 informs corrective response:

Interview with Expert #3 informs corrective response: